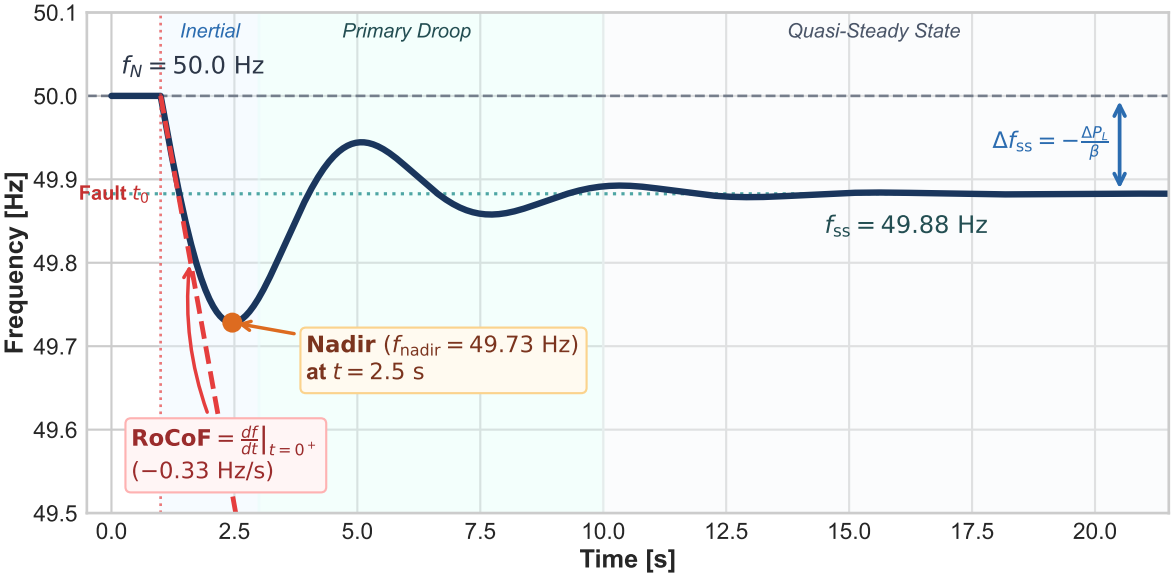
Inertial
Primary Droop
Quasi-Steady State
$f_N = 50.0$ Hz
Fault $t_0$
$\Delta f_{ss} = -\frac{\Delta P_L}{\beta}$
$f_{ss} = 49.88$ Hz
Nadir ($f_{nadir} = 49.73$ Hz) at $t = 2.5$ s
RoCoF $= \frac{df}{dt}\big|_{t=0^+}$ ($-0.33$ Hz/s)
Frequency [Hz]
Time [s]
50.1
50.0
49.9
49.8
49.7
49.6
49.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0